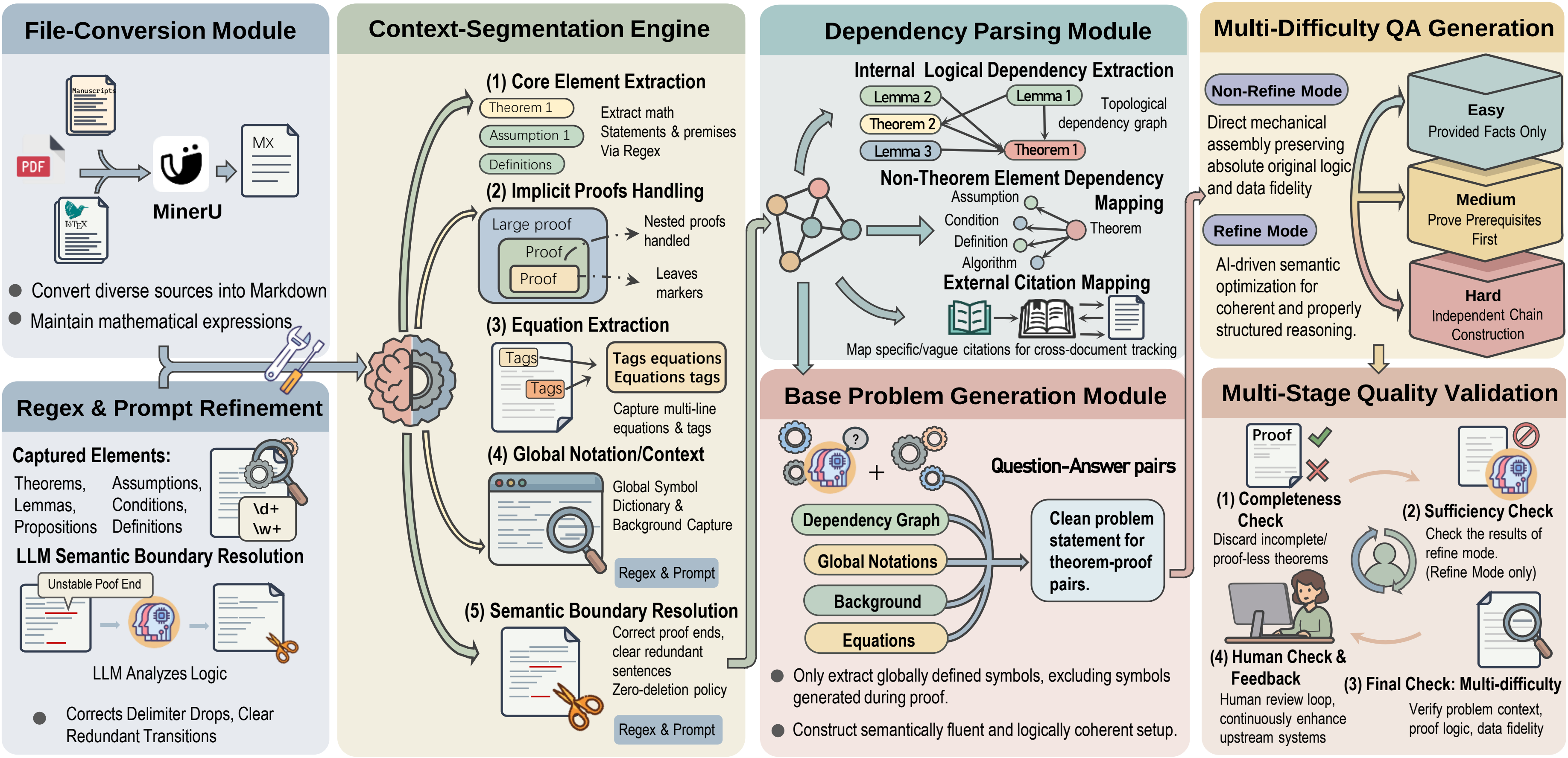

File-Conversion Module
Manuscripts
PDF
MX
MinerU
Convert diverse sources into Markdown
Maintain mathematical expressions
Regex & Prompt Refinement
Captured Elements:
Theorems, Lemmas, Propositions
Assumptions, Conditions, Definitions
\d+
\w+
LLM Semantic Boundary Resolution
Unstable Poof End
LLM Analyzes Logic
Corrects Delimiter Drops, Clear Redundant Transitions
Context-Segmentation Engine
(1) Core Element Extraction
Theorem 1
Assumption 1
Definitions
Extract math Statements & premises Via Regex
(2) Implicit Proofs Handling
Large proof
Proof
Proof
Nested proofs handled
Leaves markers
(3) Equation Extraction
Tags
Tags
Tags equations
Equations tags
Capture multi-line equations & tags
(4) Global Notation/Context
Global Symbol Dictionary & Background Capture
Regex & Prompt
(5) Semantic Boundary Resolution
Correct proof ends, clear redundant sentences
Zero-deletion policy
Regex & Prompt
Dependency Parsing Module
Internal Logical Dependency Extraction
Lemma 2
Theorem 2
Lemma 3
Lemma 1
Theorem 1
Topological dependency graph
Non-Theorem Element Dependency Mapping
Assumption
Condition
Definition
Algorithm
Theorem
External Citation Mapping
Map specific/vague citations for cross-document tracking
Base Problem Generation Module
Question-Answer pairs
Dependency Graph
Global Notations
Background
Equations
Clean problem statement for theorem-proof pairs.
Only extract globally defined symbols, excluding symbols generated during proof.
Construct semantically fluent and logically coherent setup.
Multi-Difficulty QA Generation
Non-Refine Mode
Direct mechanical assembly preserving absolute original logic and data fidelity
Refine Mode
AI-driven semantic optimization for coherent and properly structured reasoning.
Easy
Provided Facts Only
Medium
Prove Prerequisites First
Hard
Independent Chain Construction
Multi-Stage Quality Validation
Proof
(1) Completeness Check
Discard incomplete/ proof-less theorems
(2) Sufficiency Check
Check the results of refine mode. (Refine Mode only)
(3) Final Check: Multi-difficulty
Verify problem context, proof logic, data fidelity
(4) Human Check & Feedback
Human review loop, continuously enhance upstream systems